

Peter FAI

Cloud Security (Architect & Engineer)

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Professional Summary:

- With **9+** years of experience and a proven track record as a result-driven **Cloud Solutions Architect (AWS, AZURE, and GCP)** and a seasoned **Cloud Security Engineer**, I have been designing, implementing, and managing **highly scalable and secure cloud-based solutions**. I am proficient in various **AWS services**, including **EC2, S3, VPC, RDS, IAM**, and more. I have a deep understanding of **cloud architecture principles** and best practices, and I am recognized for expertise in crafting comprehensive **cloud security strategies** to **safeguard cloud environments** and ensure compliance with industry standards such as **ISO 27001** and **NIST**. My experience includes implementing **cyber security controls**, managing identity and access management (IAM), and conducting regular security risk assessments. I am skilled at translating business requirements into technical solutions, with a strong focus on threat detection and response. I am also a strong communicator and team player, and can work effectively with technical and non-technical stakeholders.

Certifications:

- AWS Certified Practitioner
- Azure DP-300/Cyber Security

Technical Skills:

- Cloud Platforms & Architecture:** AWS (Primary), Azure, GCP – Architecture Design, Security, Compliance, Multi-Cloud Integration, Google AI Platform (Vertex AI, BigQuery ML).
- Cloud Security & IAM:** Identity and Access Management (IAM), AWS Identity Center (SSO), Role-Based Access Control (RBAC), Least Privilege Enforcement, Key Management (KMS), AWS Shield, Security Hub, GuardDuty, CloudTrail, AWS Config, Control Tower, Inspector, Azure Active Directory, Microsoft Defender for Cloud Setup.
- Machine Learning & AI Services:** Model training, deployment, and monitoring on Google AI Platform, AWS SageMaker, and Azure ML; Data preprocessing, feature engineering, and ML lifecycle management.
- DevOps & Automation:** Terraform, AWS CloudFormation, CodePipeline, Jenkins, Docker, Kubernetes, CI/CD Pipelines, Infrastructure as Code (IaC).
- Monitoring & Logging:** CloudWatch, CloudTrail, AWS Config, Splunk, CrowdStrike, Azure Monitor, GCP Operations Suite (formerly Stack driver).
- Networking & Infrastructure:** VPC, Subnets, Security Groups, Direct Connect, Transit Gateway, Load Balancers, Hybrid Cloud Networking, Zero Trust Architecture.
- Cybersecurity & Forensics:** Penetration Testing, Threat Hunting, SAST/DAST, OWASP, Nessus, Metasploit, FTK, Cellebrite, SOAR Platforms, Digital Forensics, Incident Response.
- Compliance & Governance:** HIPAA, PCI-DSS, GDPR, NIST, ISO 27001 – Governance Models, Risk Assessment, and Auditing.
- Cloud Cost & Resource Management:** AWS Budgets, Cost Explorer, Savings Plans, Lifecycle Manager, FinOps Strategy for Multi-Cloud Environments.
- Cloud Migration:** AWS DMS, SMS, Snowball, Migration Hub, ADS, Azure Migrate, GCP Transfer Service.
- Programming & Tools:** Python (for ML/AI), JSON, YAML, Linux, Git, GitHub, VS Code, IntelliJ.
- Endpoint & Workload Security:** Full-stack resource protection, Security Monitoring across hybrid and multi-cloud environments, Microsoft Defender Endpoint Protection.

Professional Experience:

OneTrust|Oct 2024 - Present

Cloud Security Architect

- Designed secure **multi-account AWS landing zones** using **Control Tower, SCPs**, and **AWS Config Rules**.
- Integrated **AWS Security Hub, GuardDuty**, and **CloudTrail** for centralized **threat visibility**.
- Implemented **IAM governance** and **SSO using AWS Identity Center** with least privilege enforcement.
- Automated infrastructure provisioning using **Terraform** and **CloudFormation**.
- Led vulnerability management program leveraging **Inspector, Nessus**, and **SAST/DAST** tools.
- Conducted cloud penetration tests, forensic investigations (**FTK, Cellebrite**), and **incident response**.
- Partnered with compliance teams to ensure **PCI-DSS** and **HIPAA adherence** during audits.
- Optimized cloud spends using **Cost Explorer, Savings Plans**, and **lifecyclepolicies**.
- Azure Identity & Access Management (IAM):** Designed and implemented Azure AD security models, including RBAC, Conditional Access, MFA, and Privileged Identity Management to strengthen identity governance.
- Microsoft Defender Setup & Management:** Deployed and configured Microsoft Defender for Cloud, Defender for Endpoint, and Defender for Identity to ensure threat detection and protection across hybrid workloads.
- Azure Security Architecture:** Architected secure Azure landing zones, enforced Zero Trust principles, and implemented NSGs, Azure Firewall, and DDoS protection to safeguard workloads.
- SIEM & Threat Detection:** Integrated Microsoft Sentinel with Azure workloads, developed custom detection rules, and automated incident response playbooks using Logic Apps and SOAR.

- **Compliance & Governance:** Ensured Azure environments adhered to CIS Benchmarks, ISO 27001, NIST, HIPAA, and PCI-DSS standards by enforcing Azure Policy and Blueprints.
- **Cloud Monitoring & Logging:** Configured Azure Monitor and Log Analytics for proactive threat detection, anomaly monitoring, and operational visibility.
- **Azure Migration & Modernization:** Supported application and database migrations to Azure using Azure Migrate, Database Migration Service, and Storage Migration Service.
- Set up and managed **Microsoft Defender** for Cloud and **Azure Sentinel** for proactive threat detection and
- Deployed IAM policies, RBAC, and SSO integrations to strengthen identity security across cloud platforms.
- Leveraged **Google AI Platform** to build machine learning models for anomaly detection, improving threat detection accuracy by 30%.
- Automated compliance checks using **Azure Policy**, and GCP Security Command Center, reducing audit preparation time by 40%.

JPMorgan Chase | Jun 2023 to Sep 2024

Cloud Security Engineer

- **Security Architecture & Compliance:** I have designed and implemented secure AWS architectures, ensuring compliance with industry frameworks such as **NIST, CIS, SOC 2, and ISO 27001**. I have enforced security controls across AWS environments, aligning with regulatory requirements such as PCI-DSS and HIPAA.
- **Threat Model Development, Security Risk Assessment:** Develop and maintain threat models for cloud environments (AWS, Azure, GCP) using the MITRE ATT&CK framework to identify potential attack vectors and risks specific to cloud infrastructure.
- Designs and implements security automation, determines requirements, specifications, and designs of cyber security solutions, and assists in major system security enhancements.
- **Implementation of Security Controls:** Based on findings from threat models, recommend and implement cloud security controls to mitigate identified risks and enhance the overall security posture.
- **Threat Detection & Incident Response** – I have integrated AWS security services with SIEM solutions like Splunk, enabling real-time threat
- **IAM & Zero Trust Security:** I have designed and enforced least privilege access controls in AWS using **IAM roles**, policies, and AWS Organizations SCPs. I have implemented multi-factor authentication (MFA), RBAC, and Zero Trust security models to enhance identity and access management.
- Managed and **configured AzureAD**, leveraging Conditional Access policies and Entra ID Government Platform for enhanced security and compliance in government environments.
- Implemented and maintained Role-Based Access Control (RBAC) to ensure appropriate access management and governance across cloud resources.
- Remediated **security-related vulnerabilities** across collaboration workloads, proactively addressing potential risks and ensuring the integrity of cloud-based services.
- Empower personnel with essential cyber security knowledge and skills through **cyber security** awareness training sessions tailored to their roles and responsibilities.
- **Collaborate with Security Teams:** Work closely with **SOC, DevOps**, and **cloud engineering** teams to ensure that threat models are effectively integrated into cloud security monitoring and incident response workflows.
- **DevSecOps & CI/CD Security:** I have embedded security controls into **CI/CD pipelines**, ensuring secure code deployments using AWS Code Pipeline, GitHub Actions, and container security tools for **EKS and ECS**. I have worked closely with **DevOps** teams to integrate security into cloud-native application development.
- **Cloud Security Operations:** I have hands-on experience managing **AWS security** services, including **AWS IAM, Guard Duty, Security Hub, Macie, AWS WAF, and KMS**. I have monitored security events, enforced security policies, and conducted security posture assessments across AWS environments.
- **Security Audits and Compliance:** Support security audits by providing insights from threat models and ensuring cloud environments meet compliance requirements such as CIS Benchmarks, **ISO 27001**, and **GDPR**.
- **Infrastructure Security & Automation:** I have implemented security best practices in Infrastructure as Code (IaC) using Terraform and AWS CloudFormation, ensuring consistent security controls across cloud deployments
- **Automation of Threat Detection:** Develop and automate threat detection rules and alerts within cloud environments by mapping cloud logs and telemetry data to **ATT&CK** techniques.
- **Documentation and Reporting:** Create detailed documentation of threat models, security controls, and mitigation strategies, presenting them to leadership for risk management and security investment decisions.
- Responded promptly to customer requests related to cyber security issues, providing solutions and recommendations for risk mitigation.
- **Application Security (AppSec). Vulnerability Management:** Identifying and addressing security risks through threat modeling, secure coding, and testing methods (**SAST, DAST, pentesting**).

Cogent Infotech | Oct 2022 to Jun 2023

CLOUD SECURITY ENGINEER

- **Cloud Security Architecture Design and Proof of Concept Implementations.** I designed and implemented **secure cloud architectures**, incorporating best practices in identity and access management (**IAM**), encryption, and **network security** to protect sensitive data and ensure robust cloud environments. I developed proof-of-concept implementations to validate **architectural decisions**, evaluate the functionality of new **cloud services**, and demonstrate proposed solutions to **stakeholders**.
- **Identity and Access Management (IAM):** I established and maintained IAM policies, enforced the principle of least privilege, and implemented **multi-factor authentication (MFA)** and single sign-on (**SSO**) solutions to strengthen access controls.

- **Security Operations Center (SOC) Leadership:** Established and efficiently managed a Security Operations Center (SOC) to strengthen overall **cyber security posture**. Collaborated with cross-functional teams to enhance security operations and incident response strategies.
- **Data Encryption and Network Security:** I implemented end-to-end encryption strategies using services like AWS Key Management Service (KMS) and Azure Key Vault to secure data at rest, in transit, and during processing. Additionally, I configured security groups, **network ACLs**, and **Virtual Private Clouds (VPCs)** to enforce secure network architectures. Incident Response and Vulnerability Management: I led incident response efforts, developed and tested incident response plans, and conducted forensic analyses to identify root causes of incidents. I also established a vulnerability management program, performed regular scans, and collaborated with teams to remediate identified vulnerabilities.
- **Security Compliance and Automation:** I ensured compliance with industry standards and regulations such as **GDPR**, **HIPAA**, and **PCI DSS** through audits and control implementations. Additionally, I developed security automation scripts to enforce policies, monitor events, and integrate security checks into CI/CD pipelines.
- **Threat Intelligence and Continuous Monitoring:** I stayed updated on the latest cybersecurity threats, incorporated **threat intelligence** into strategies, and **implemented threat** detection mechanisms. I deployed continuous monitoring solutions using SIEM tools and log analysis to detect and respond to security events in real-time.

L&T| May 2020 - Jun 2022

CLOUD ENGINEER

- Collaborated with cross-functional teams to design, implement, and manage cloud infrastructure solutions, leveraging **AWS**, **Azure**, and **Google Cloud Platform**.
- Implemented infrastructure-as-code using **Terraform**, **CloudFormation**, and **Azure Resource Manager**, ensuring consistent and repeatable deployments across environments.
- Orchestrated containerization technologies like **Docker** and **Kubernetes**, facilitating efficient application deployment and management.
- Collaboration Worked closely with Microsoft security researchers and utilized a suite of Microsoft security products such as **Office 365 Security**, **MCAS**, **WDATP**, **ATP**, **MDI**, and Research Studio to enhance cybersecurity measures.
- Ensured the security and integrity of cloud infrastructure by **implementing networking concepts, including VPNs, load balancers, and firewalls**.
- Collaborated closely with teams to establish and **enforce monitoring and logging practices, ensuring system performance** and availability met organizational standards.
- Proactively Troubleshooted and resolved issues to maintain optimal system functionality, minimizing downtime and disruptions.
- Enhanced **cyber security capabilities** by **gaining experience in log management** and **Wi-Fi penetration testing**. Applied security best practices to ensure continuous monitoring and quick detection of **potential vulnerabilities**.
- Developed automation scripts **using Python, Bash, and PowerShell** to streamline processes and enhance operational efficiency.
- Designed and implemented robust **networking solutions, including VPNs, load balancers, and firewalls** to ensure secure cloud environments.

TDCJ, Texas| Aug 2016 to Feb 2020

CLOUD SOLUTIONS ARCHITECT

- Led the design and **architecture of 20+ AWS cloud solutions**, ensuring they met performance, security, and scalability requirements.
- Reduced infrastructure costs by **15%** through the optimization of AWS services and resources.
- Established and maintained **robust security controls**, resulting in zero security breaches during the **2.5 years** in the role.
- Developed and executed migration strategies for **10+ applications**, minimizing downtime and achieving a **30% reduction** in operational costs.
- Led third-party risk assessments, providing detailed security evaluations and recommending improvements for **vendor management processes** within NYC OTI's **cyber security** framework.
- Implemented **AWS Cost Explorer** and **budget alerts**, leading to a **20%** reduction in monthly cloud expenditure.
- Engage with clients through various channels, **including meetings, emails, and phone calls**, to understand their needs and concerns, providing tailored solutions and support. Ensure timely and effective communication with **customers, addressing inquiries** and feedback promptly, and striving to exceed expectations by delivering exceptional service with empathy and professionalism.
- Gather customer feedback to identify areas for improvement, collaborating with internal teams to enhance products, services, and processes based on client **insights, ultimately fostering** strong relationships and driving **customer satisfaction**.
- Automated infrastructure scaling with **AWS Lambda** and **CloudWatch**, improving application performance during traffic spikes.
- Achieved a disaster **recovery RTO** of 2 hours and an **RPO of 15 minutes**, ensuring data integrity and availability.
- Configured **AWS Transit Gateway** to optimize network performance, resulting in a **25% reduction** in latency for inter-VPC communication.
- **Deployed Amazon Route 53** for global load balancing, enhancing **web service availability** and **resilience**.

Educations:

- **Greenville University**, Bachelor's degree, Information Technology
- **University of Buea**, Bachelor of Business Administration - BBA